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Bi-Level Multi-Criteria Optimization to Include Linear Energy Transfer Into Proton Treatment Planning

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ABSTRACT

In proton therapy treatment planning, the aim is to ensure tumour control while sparing the various surrounding risk structures. The biological effect of the irradiation depends on both physical dose and linear energy transfer (LET). In order to include LET alongside physical dose in plan creation, we propose to formulate the proton treatment planning problem as a particularly structured multi-criteria bi-level optimization problem, which we call hierarchical. We show that the hierarchical multi-criteria bi-level problem can be reduced to a standard multi-criteria optimization (MCO) problem employing a specific domination cone. As the unfavourable properties of this cone prohibit a direct application of standard MCO solution methods, we further illustrate how a more convenient approximate cone can be constructed. Based on the found reduction to a standard MCO problem, we then describe a novel approach to calculate a representation of the non-dominated points for the hierarchical multi-criteria bi-level problem. As a point of reference, we also discuss a second, more brute-force approach. We apply both approaches to a prostate and a head and neck case and compare the results.

1 | Introduction

In external radiation therapy, a target area is irradiated from the outside in an attempt to destroy cancerous cells through radiation-induced damage. At the same time, nearby risk structures must be spared as much as possible in order to avoid side effects. Finding the best balance between tumour control and risk sparing motivates the application of multi-criteria optimization (MCO) methods. Over the last two decades, MCO methodology has been successfully established for both photon and proton treatment planning (Küfer et al. 2002, 2003; Monz et al. 2008; Craft et al. 2012; Breedveld et al. 2019).

As outlined in the review paper (Breedveld et al. 2019), *cost-based* strategies for solving the multi-criteria treatment planning problem use *cost functions* that measure desirable properties of a treatment plan. A cost function can be a dose metric, such as

the mean or maximum dose to a particular organ, or an additional plan characteristic such as treatment time. Two distinct approaches are discussed in Breedveld et al. (2019): automated and interactive cost-based MCO. In automated cost-based MCO, the preferences of the decision-maker are encoded in a wish-list, and an algorithm determines the best plan according to those stated preferences. In contrast, the interactive approach does not require a priori formulation of the decision maker's preferences. Rather, in a pre-computational first step, a set of Pareto efficient plans is calculated that covers the range of clinically feasible treatment options (*plan creation*). Subsequently, the plan set can be interactively searched by the decision-maker for the most preferable compromise (*navigation*).

Compared to photons, proton radiation is more complex, resulting in a less straightforward relationship between physical dose and biological effect. One way to more accurately reflect the

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biological effect of protons is to integrate linear energy transfer (LET) into the plan optimization, as higher LET leads to increased biological effectiveness (Paganetti and van Luijk 2013; Chaudhary et al. 2014). A straightforward method to integrate LET into plan optimization—which we will apply as well—is to include objectives that evaluate the product LETxD of LET and physical dose (Unkelbach et al. 2016; Bai et al. 2020; Gu et al. 2021).

A complete integration of LET into multi-criteria proton treatment plan optimization and decision support without compromising the dose distribution has not yet been described. In Giantsoudi et al. (2013), the authors depict a way to retroactively include LET into the decision making for an already optimized plan set, but do not consider LET in the plan creation.

In this work, we discuss how to properly include LET re-optimization into the plan generation step of a cost-based interactive multi-criteria proton treatment plan optimization approach. LET objectives can be included into proton therapy planning alongside the dose objectives in a one-step optimization problem or as a second step after dose optimization. Currently, there is no evidence whether one type of model is preferable over the other (McIntyre et al. 2023). However, while the use of MCO with one-step methods is straight forward, two-step or re-optimization methods pose challenges in multi-criteria settings, for example when it comes to ensuring the representatively calculated non-dominated points are a good reflection of the whole set of interest, or when it comes to the question how to handle interactive decision making. Our work aims to help close this gap by presenting a method to handle such two-step problems in a single optimization step. Using re-optimization reflects a treatment planner who prioritises dose objectives and wished to improve LET only if the trade-off to the dose objectives is not too big. To account for this kind of preference, we propose to use bi-level optimization as a mathematical framework. Let F denote a vector-valued objective measuring the plan quality in terms of physical dose, and let ϕ denote a vector-valued objective measuring the plan quality in terms of LETxD. Further, let G and ψ be vector valued functions describing the constraints. Then the bi-level problem

$$\begin{aligned} \min_{x_u, x_l} \quad & F(x_u) \\ \text{s. t.} \quad & G(x_u) \leq 0, \\ & x_l \in \operatorname{argmin}_{x_l} \{ \phi_{x_u}(x_l) \text{ s. t. } \psi_{x_u}(x_l) \leq 0 \} \end{aligned} \quad (1)$$

reflects our aim to improve LETxD without sacrificing aspects of the physical dose distribution. Note that if the lower level problem has a feasible solution independent of x_u , and both F and ϕ were real-valued, solving (1) would be equivalent to a lexicographic approach. In the multi-criteria setting however, the problem has not been extensively studied.

In a previous work, Unkelbach et al. treated the optimization of physical dose and LETxD as a two-step problem (Unkelbach et al. 2016). After finding a desirable dose distribution, the achieved objective values were incorporated into the constraints and—in a second step—LETxD in the organs at risk

was minimized. However, a fixed scalarization was employed in both steps, resulting in a straightforward lexicographic procedure. In contrast, we integrate dose and LETxD objectives into a unitary multi-criteria problem formulation and then find a representation of the whole non-dominated set. As a comparison, we also calculate solutions step-wise by calculating a representative set of the solutions of the dose-based MCO problem and then reoptimizing each of these solutions with respect to the LETxD objectives. We refer to this approach as the step-wise approach. Due to the solving of two MCO problems this approach does not offer a guarantee of how well the set of interesting solutions is represented.

Our conceptual idea to solve (1) is the following: Find a domination cone and the corresponding induced order relation such that the solution set of a standard multi-criteria optimization problem under this ordering coincides with the solution set of the original bi-level problem. Then solve this standard MCO problem using established methods to approximate the non-dominated set. In our case, the bi-level problem (1) is a multi-criteria optimization problem in both levels, but it has additional properties that are conducive to such a reformulation.

The first ones to follow such an approach were Fliege and Vicente (2006). They reformulated specific bi-level problems—namely those with one objective in each level, and where the upper level constraints are independent of the lower level variables—to MCO problems. They described an order relation that ensured the equivalence of solutions between the two problem formulations. To find a fitting cone, they approximated the original order relation with a weaker one such that the set of non-dominated points with regard to this weaker relation was a subset of the original non-dominated points. One objective of the multi-criteria problem they constructed was the gradient of the lower level objective with respect to the lower level variable, aimed at achieving a minimum in the lower-level problem. This implied that the equality of the solutions between both formulations occurs only when the lower level problem is convex for all fixed upper level values. Otherwise, not all solutions of the bi-level problem can be obtained as solutions of the MCO problem. They introduced a cone that induces the desired order relation for the constructed MCO problem. However, their cone was not convex, and the dual cone only contained $\{0\}$, making it unfit to use in a weighted sum approach. The cone was the union of two convex cones, so they instead took the minimum between two scalarizations, one for each cone. Linear scalarizations were not sufficient and they instead proposed to use a quadratic mapping. However, the scalarization and its parameters are problem dependent, and finding them is not a trivial task.

In Ivanenko and Plyasunov (2008) the authors extended the ideas from Fliege and Vicente (2006) to problems where the upper level constraints depend on the lower level variables. They considered bi-level problems with a single objective in each level; thus the results are not directly applicable to our problem. They focused on relations and did not construct a cone, nor specified how the relation can be used for numerical optimization.

Ruuska et al. (2012) also generalised the approach from Fliege and Vicente (2006). They considered multi-criteria problems

at both levels, again with the restriction that the upper level constraint does not depend on the lower level variable. They obtained a relation and a standard MCO problem whose efficient solutions coincide with those of the original bi-level problem. However, the obtained relation was not transitive. Moreover, the relation contains the definition of efficiency, and in the worst case, every evaluation of whether a point is smaller than another requires solving an MCO problem. To circumvent this, they proposed a weaker relation instead and showed that there are cases where this weaker relation is transitive, but this does not hold for the general case. The MCO problem they used with these relations has dimensions equal to the sum of the dimensions of the lower and upper level objective spaces added to the dimension of the upper level decision space, making it high dimensional. While they only used the standard domination cone for the bilevel problem, their approach should also work for arbitrary domination cones. A generating cone was not found, and they did not elaborate on how to efficiently use the relation in numerical optimization routines.

As a special case of the above, LET reoptimization motivates us to investigate bi-level problems where the upper level problem is independent of the lower level variable, and the feasible set of the lower level problem depends only on the upper level function value, not the upper level variable itself. We will show that this special structure enables us to find a particular order relation and a generating cone to identify the solutions of the bi-level problem with those of a single-level MCO problem. The resulting MCO problem is only of the dimension of both objective spaces from the upper and lower level problem. Additionally, the problem structure allows us to define an approximate cone that is polyhedral and whose dual cone has non-empty interior. These two properties can be exploited in numerical optimization routines, as we will demonstrate. The reformulation to a single-level MCO allows us to use known algorithms from MCO which guarantee a good approximation of the non-dominated set.

Our paper is organised as follows. In the theoretical part of our paper (Section 2), we formally introduce this particular type of bi-level problem and investigate its theoretical properties. In particular, we show how the problem can be reformulated into a single-level MCO problem by defining a specific domination cone. We then discuss two algorithmic approaches to solve the problem. The first algorithm draws on the approach from Unkelbach et al. (2016) for proton treatment planning, generalising it to the multi-criteria setting. The second algorithm exploits the aforementioned reformulation. In the results part (Section 3) we apply both algorithms to two real-world proton planning problems and compare the results.

2 | Materials and Methods

2.1 | Terms and Definitions

First, we will introduce some general notations and definitions that will be used throughout the paper. The interior of a set S is denoted by $\text{int}(S)$, its boundary by $\text{bnd}(S)$. For two sets S_1 and

S_2 , $S_1 + S_2$ refers to their Minkowski sum, and for a point s and a set S the sum $s + S$ is the Minkowski sum between $\{s\}$ and S . Further, the dual of a cone C is denoted by C^* .

The following Definitions 2.1–2.4 as well as Lemma 1 introduce fundamental concepts from the theory of MCO.

Definition 2.1. A multi-criteria optimization problem is an optimization problem with a vector valued objective function. It can be written as

$$\begin{array}{ll} \min & F(x) = (F_1(x), \dots, F_n(x)) \\ \text{s. t.} & x \in X \end{array} \quad (\text{MCO})$$

where $X \subseteq \mathbb{R}^m$ is the feasible set and $F: X \rightarrow \mathbb{R}^n$.

If the objective functions $F_1, \dots, F_n$ and the feasible set X are convex, the MCO problem is convex. We denote the image of F as $F(X) = \mathcal{Y}$. Typically, not all objective functions can take their minimal possible value at the same time. Instead, the best possible compromises between the objectives are considered as solutions of the problem.

To define minima with respect to the vector-valued evaluation F , an order relation on $\mathcal{Y}$ must be defined that determines whether a vector is smaller than another. We use sets to define such relations. The relation induced by the set S is

$$x \leq_S y \Leftrightarrow y - x \in S.$$

An order relation is a relation that is reflexive, anti-symmetric, and transitive. A convex and pointed cone C that contains 0 is called proper and induces an order relation. The cone is referred to as the domination cone. A prominent domination cone is $\mathbb{R}_{\geq 0}^n$, also called the standard domination cone. For this cone $x \leq_{\mathbb{R}_{\geq 0}^n} y \Leftrightarrow x_i \leq y_i \quad \forall i = 1, \dots, n$. If $\mathbb{R}_{\geq 0}^n$ is employed, we denote the induced order relation simply as $x \leq y$. The concept can also be applied to matrices, where the relation is understood element-wise as well.

The following definition explains when a solution is considered the best possible compromise.

Definition 2.2. Ehrgott (2005) (Def 1.23, Def 2.1) A point $x \in X$ is called efficient if there is no other $\tilde{x} \in X$ such that $F(\tilde{x}) \leq_{C \setminus \{0\}} F(x)$. The point $F(x) \in \mathcal{Y}$ is called non-dominated. The set of non-dominated points is

$$P_C = \{F(x) | x \in X, \nexists \tilde{x} \in X: F(\tilde{x}) \leq_{C \setminus \{0\}} F(x)\} \quad (2)$$

and is referred to as front. The set of efficient points is denoted as E_C . For $C = \mathbb{R}_{\geq 0}^n$, the terms Pareto efficient and Pareto front are used.

We will write $(C\text{-min})$ to refer to the operator for the non-dominated points of an MCO problem with respect to a cone C . For the standard domination cone we will use $\min$. A non-dominated point with respect to $\mathbb{R}_{\geq 0}^n$ can be understood as a

point such that no other point is equally good or better in all objectives. A weakly non- $\mathbb{R}_{\geq 0}^n$ -dominated point is one where no other point is truly better in all objectives.

Definition 2.3. Ehrgott (2005) (Def 2.24) A point $x \in X$ is weakly efficient if there is no other point $\tilde{x} \in X$ such that $F(\tilde{x}) \leq_{\text{int}(C)} F(x)$. The respective point $F(x) \in \mathcal{Y}$ is weakly non-dominated.

One way to find efficient solutions of an MCO problem is to scalarize it, i.e., to transform it into a parameterized standard optimization problem with a real-valued objective. The scalarized problem can then be solved with standard optimization methods such as gradient descent.

Definition 2.4. The weighted sum scalarization of (MCO) is

$$\begin{aligned} \min \quad & \sum_{i=1}^n w_i F_i(x) \\ \text{s. t.} \quad & x \in X \end{aligned} \quad (\text{WS})$$

with $w \in \mathbb{R}^n \setminus \{0\}$.

If the weights for (WS) are chosen from the dual cone C^* , one can solve (WS) to obtain a (weakly) non-dominated point of (MCO). The proof for the following lemma from Serna (2012) can be found in Appendix A.

Lemma 1. Serna (2012) (Lemma 1.23) If the weights w of (WS) are from $C^* \setminus \{0\}$ and $\sum_{i=1}^n w_i F_i(x^*) = \min \{ \sum_{i=1}^n w_i F_i(x) \mid x \in X \}$, then $F(x^*)$ is a weakly non-dominated point of (MCO) with regard to C . If $w \in \text{int}(C^*)$, $F(x^*)$ is a non-dominated point.

Since commonly the front cannot be calculated explicitly, some non-dominated points are calculated and used to approximate the whole front.

Definition 2.5. A set $Y^O \subseteq \mathbb{R}^n$ is an outer approximation of the set $Y \subseteq \mathbb{R}^n$ with respect to the cone C if for any $y \in Y$ there is a $\hat{y} \in Y^O$ such that $\hat{y} \leq_C y$ and if for all $y^O \in Y^O$ it holds that $y^O \notin Y + \text{int}(C)$.

Definition 2.6. For a set $Y \subseteq \mathbb{R}^n$, the set $Y^I \subseteq Y + C$ is an inner approximation with respect to cone C if for all $y \in Y$ there is a $\hat{y} \in Y^I$ such that $y \leq_C \hat{y}$.

This definitions imply $Y^O + C \supseteq Y + C \supseteq Y^I + C$, which is the intuition of the inner and outer approximation.

Algorithms that use an inner and outer approximation are called sandwiching algorithms. Their aim is to find an approximation with as few solves as possible. For this, they calculate the next weight to be used for (WS) such that the newly calculated point reduces the maximal approximation error as much as possible. For a pseudo-code formulation of the Sandwiching algorithm we employed, see Bokrantz and Forsgren (2012, Algorithm 3.1). Assuming the domination cone C contains $\mathbb{R}_{\geq 0}^n$, it is designed to handle convex MCO problems and uses the convex hull of the calculated non-dominated points as inner approximation and the

intersection of the positive half spaces of the hyperplanes supporting these non-dominated points as outer approximation of $P_C + C$. Since the true front is in between the inner and outer approximation, their distance can be used as upper threshold of the approximation error.

In the remainder of this section, we introduce the bi-level problem as a special kind of optimization problem. An optimization problem is a bi-level problem if the constraints of the so-called upper level problem include an optimization problem, the so-called lower level problem. In particular, the upper level optimization variable x_u is a parameter in the lower level problem. We henceforth denote a parameter x_l influencing a function ϕ as $\phi_{x_l}(x_u)$ and a variable as $\phi(x_u, x_l)$.

Definition 2.7. A bi-level problem can be formulated as

$$\begin{aligned} \min_{x_u, x_l} \quad & F(x_u, x_l) \\ \text{s. t.} \quad & x_l \in \underset{x_l}{\text{argmin}} \{ \phi_{x_u}(x_l) : \psi_{x_u}(x_l) \leq 0 \}, \\ & G(x_u, x_l) \leq 0. \end{aligned} \quad (\text{BLP})$$

A bi-level problem can be a multi-criteria problem in either level. This means ϕ or F can be vector-valued. Then finding a non-dominated point requires two decisions, one for the lower level and one for the upper level.

As motivated in the introduction, our aim is to solve a particular multi-criteria bi-level problem with certain special properties. Firstly, the upper level objectives and constraints are independent of x_l , which is only included as part of the lower level problem. Secondly, the lower level problem is only dependent on the upper level objectives values, not on the variables themselves. We call bi-level problems with these properties hierarchical. We will refer to the domination cone for its upper level as C_U and for the one of its lower level as C_L . A hierarchical bi-level problem can be formalised as:

$$\begin{aligned} \left(\begin{array}{l} C_U\text{-min} \\ x_u, x_l \end{array} \right) \quad & F(x_u) \\ \text{s. t.} \quad & G(x_u) \leq 0, \\ & x_l \in \left(C_L\text{-argmin}_{x_l} \right) \left\{ \phi_{F(x_u)}(x_l) \mid \text{s. t. } \psi_{F(x_u)}(x_l) \leq 0 \right\}. \end{aligned} \quad (\text{HBP})$$

2.2 | Theoretical Properties of the Hierarchical Bi-Level Problem

Following the concept originally introduced in Fliege and Vicente (2006) and discussed in the introduction, we reduce the hierarchical bi-level problem (HBP) to a standard MCO problem by finding a suitable domination cone. In the following, we introduce such a problem reformulation, and we show that all efficient solutions of this reformulation can be identified with an efficient solution of (HBP) and the corresponding solution of the lower level problem. We call such a reformulation a *single level reduction* (SLR) of the hierarchical bi-level problem. The single level reduction uses the objectives and constraints from both levels of (HBP) as congenial objectives and constraints, respectively.

Definition 2.8. The single level reduction of (HBP) is

$$\begin{aligned} & \left(C_{SLR} - \min_{x_u, x_l} \right) \quad F(x_u), \phi(F(x_u), x_l) \\ & \text{s. t.} \quad G(x_u) \leq 0, \\ & \quad \psi(F(x_u), x_l) \leq 0. \end{aligned} \quad (SLR)$$

To achieve an identification of the efficient points of (HBP) and (SLR) we define the cone for the single level reduction as follows.

Definition 2.9. The single level reduction cone is

$$C_{SLR}(C_U, C_L) = \{(u, l) \in \mathbb{R}^{n_U+n_L} \mid u \in C_U \setminus \{0\} \vee (u = 0, l \in C_L)\}.$$

If the used cones are clear, we will simply write C_{SLR} . The properties required for a domination cone to ensure that it induces an order relation are transferred from C_U and C_L to C_{SLR} .

Lemma 2. 1. If C_L contains 0, then $C_{SLR}(C_U, C_L)$ contains 0.

2. If C_U and C_L are pointed, then $C_{SLR}(C_U, C_L)$ is also pointed.

3. If $C_U \setminus \{0\}$ and C_L are convex cones, then $C_{SLR}(C_U, C_L)$ is also convex.

4. If a cone C is pointed and convex, $C \setminus \{0\}$ is convex.

5. If C_U and C_L are proper cones, $C_{SLR}(C_U, C_L)$ is proper.

Proof. 1. C_{SLR} includes all points that are zero for the dimensions corresponding to the upper level and have arbitrary values for $l \in C_L$. Thus, if $0 \in C_L$, then $(0, 0)^T \in C_{SLR}$.

2. Assume $d = (d_u, d_l)^T \in C_{SLR}$. Then $-d = (-d_u, -d_l)^T$. If there is an i such that $d_{u_i} \neq 0$, then $-d_{u_i} \notin C_U$ and consequently $-d \notin C_{SLR}$. If $d_u = 0$ and there is a j such that $d_{l_j} \neq 0$, then $-d_{l_j} \notin C_L$ and consequently $-d \notin C_{SLR}$.

3. By assumption, $C_U \setminus \{0\}$ and C_L are convex. Let $a, b \in C_{SLR}(C_U, C_L)$. Clearly, if $a_u, b_u \in C_U \setminus \{0\}$, so is their sum and if $a_u = b_u = 0$ and $a_l, b_l \in C_L$, so is their sum. Without loss of generality, assume $a_u \in C_U \setminus \{0\}$, $b_u = 0$, and $b_l \in C_L$. Then $a_u + b_u = a_u + 0 = a_u \in C_U \setminus \{0\}$. Hence, $a + b \in C_{SLR}(C_U, C_L)$.

4. Since C is convex, for any $a, b \in C$ it follows that $a + b \in C$. Assume $C \setminus \{0\}$ is not convex. Thus, there are two points $a, b \in C \setminus \{0\}$ such that $a + b \in C$ but $a + b \notin C \setminus \{0\}$. Hence, $a + b = 0$. Then, $a = -b$ and because C is pointed consequently $a = b = 0$. This contradicts $a, b \in C \setminus \{0\}$. Thus, $C \setminus \{0\}$ has to be convex.

5. This follows directly from 1 to 4. $\square$

Henceforth, it will be assumed that both C_U and C_L are proper. Thus, C_{SLR} is also a proper cone and consequently (SLR) a well-defined MCO problem. The following theorem shows how the efficient set of (HBP) with regard to C_U and C_L is connected to the efficient set of (SLR) with regard to $C_{SLR}(C_U, C_L)$.

Theorem 2.1. If for (HBP), every C_U -efficient point of the upper level leads to a lower level problem with a well-defined non-empty solution, then the efficient sets of (HBP) and (SLR) are the same, i.e.,

$$E_{C_U}(HBP) = E_{C_{SLR}}(SLR).$$

Proof. According to Lemma 2, all considered cones are proper. Thus, they define order relations that can be used for the definition of the respective efficient sets:

$$E_{C_U}(HBP) = \left\{ \begin{pmatrix} x_u \\ x_l \end{pmatrix} \mid G(x_u) \leq 0, \psi(F(x_u), x_l) \leq 0, \right. \quad (3a)$$

$$\nexists \hat{y}_l: \phi_{F(x_u)}(\hat{y}_l) \leq_{C_L \setminus \{0\}} \phi_{F(x_u)}(x_l), \psi_{F(x_u)}(\hat{y}_l) \leq 0, \quad (3b)$$

$$\nexists \begin{pmatrix} y_u \\ y_l \end{pmatrix}: G(y_u) \leq 0, \quad (3c)$$

$$F(y_u) \leq_{C_U \setminus \{0\}} F(x_u), \quad (3d)$$

$$y_l \in \left(C_L - \argmin_{y_l} \{ \phi_{F(y_u)}(y_l) \mid \psi_{F(y_u)}(y_l) \leq 0 \} \right), \quad (3e)$$

and

$$E_{C_{SLR}}(SLR) = \left\{ \begin{pmatrix} x_u \\ x_l \end{pmatrix} \mid G(x_u) \leq 0, \psi(F(x_u), x_l) \leq 0, \right. \quad (4a)$$

$$\nexists \begin{pmatrix} y_u \\ y_l \end{pmatrix}: G(y_u) \leq 0, \psi(F(y_u), y_l) \leq 0, \quad (4b)$$

$$[(F(y_u) \leq_{C_U \setminus \{0\}} F(x_u))] \quad (4c)$$

$$\vee (F(y_u) = F(x_u), \phi(F(y_u), y_l) \leq_{C_L \setminus \{0\}} \phi(F(x_u), x_l))]. \quad (4d)$$

We begin by demonstrating that $E_{C_{SLR}}(SLR) \subseteq E_{C_U}(HBP)$.

Let $(x_u^*, x_l^*)^T$ be a point in $E_{C_{SLR}}(SLR)$. According to (4a), x_u^* satisfies the constraints of the upper-level problem, that is, (3a). There cannot exist any $(y_u, y_l)^T$ such that the inequalities outlined in (4b) and (4c) are satisfied. This implies that there exists no point such that (3c–3e) hold. Additionally, no $(y_u, y_l)^T$, including $(x_u, y_l)^T$, can satisfy the inequalities of (4b) and (4d). This implies (3b).

Next, we show $E_{C_U}(HBP) \subseteq E_{C_{SLR}}(SLR)$.

Assume $(x_u^*, x_l^*)^T \in E_{C_U}(HBP)$. Then (3a) directly implies (4a). Furthermore, for any $(y_u, y_l)^T \in \mathbb{R}^{n_u+n_l}$, one of the inequalities described in (3c–3e) must be violated. A point violating (3c) also violates (4b). If (3d) is violated, it transfers to (4c). Since (4c) and (4d) are connected by an or condition, (4d) needs to be excluded at the same time. This is guaranteed by condition (3b). The remaining possibility is a point $(y_u, y_l)^T$ violating (3e). Such a point is either infeasible, which directly transfers to (4b), or dominated. By assumption, the lower level problem has a non-empty set of solutions for every efficient point of the upper level problem as fixed parameter. The latter implies the existence of $\hat{y}_l \in (C_L - \text{argmin}_z) \{ \phi_{F(y_u)}(z) | \psi_{F(y_u)}(z) \leq 0 \}$, which would contradict the assumption that there is no point $(y_u, y_l)^T$ such that (3c–3e) hold. Therefore, this case cannot occur. $\square$

This theorem relies on the specific structure of the hierarchical bi-level problem. In particular, the inclusion $E_{C_U}(HBP) \subseteq E_{C_{SLR}}(SLR)$ requires that the upper level variables occur in the lower level problem only via the evaluation of the upper level objectives. Otherwise, if an upper-level non-dominated point is generated by more than one efficient upper level solution, then $E_{C_U}(HBP)$ includes all combinations of upper level efficient points and respective lower level efficient points, even when two points have the same upper level value and one dominates the other in the lower level. However, in $P_{C_{SLR}}(SLR)$ among points with same upper level values only those with best lower level values are included. The independence required in (HBP) prevents this case from happening. Efficient points of the upper level that belong to the same non-dominated point will result in the same non-dominated points in the lower level.

We further required that the lower level problem has a solution for each efficient point of the upper level problem. If this is not the case, P_{HBP} would include the points with the best upper level solutions among those, where a lower level solution exists. $P_{C_{SLR}}(SLR)$ on the other hand would consist of points that are non-dominated in the upper level and have a lower level solution.

The way we include LET into the proton planning problem can be considered a reoptimization problem. The motivation of reoptimization problems is to first optimize objectives with the highest priority. Then, in the reoptimization step additional objectives are optimized while limiting the deviation of the objective values that were obtained in the first optimization. The broad class of such reoptimization problems can be expressed as a hierarchical bi-level problem. The lower level then typically contains a constraint of the form

$$F(x_l) \leq_{C_U} F(x_u) + \epsilon,$$

where ϵ is a non-negative vector describing the allowed deviation from the initial solution. Since in (HBP) and (SLR) both, x_u and x_l , are variables, this constraint is seldomly convex with respect to x_u and x_l , even if $F(x)$ is convex with respect to x . However, if ϵ is set to zero, the SLR-cone allows to remove this constraint without altering the non-dominated set.

Definition 2.10. For a hierarchical bi-level problem where the constraints of the lower level problem limit the deviation

from the objective values of the upper level problem to zero, include the constraints of the upper level problem $G(x_l) \leq 0$ and can include additional constraints $g(x_l) \leq 0$, that is,

$$\begin{aligned} F(x_l) &\leq_{C_U} F(x_u), \\ G(x_l) &\leq 0, \\ g(x_l) &\leq 0, \end{aligned}$$

the single level single variable reduction (SLSVR) is

$$\begin{aligned} & (C_{SLR} - \min_x) \quad F(x), \phi(F(x), x) \\ \text{s. t.} \quad & G(x) \leq 0, \\ & g(x) \leq 0. \end{aligned} \quad (\text{SLSVR})$$

This reduction uses that $F(x_l) \leq_{C_U} F(x_u)$ for a C_U -efficient point x_u of the upper level can only hold if $F(x_l) = F(x_u)$. The reduction of the decision space is beneficial for calculations, and does not lead to a loss of relevant information.

Theorem 2.2. Let y be an C_{SLR} -efficient point of (SLSVR). Then, (y, y) is an C_U -efficient point of the corresponding (HBP). Moreover, if (x, y) is an C_U -efficient point of (HBP) with the required constraints as per Definition 2.10, then y is an C_{SLR} -efficient point of the corresponding (SLSVR).

Proof. First, let (x, y) be an efficient point of (HBP). This implies $g(y) \leq 0$ and $G(y) \leq 0$. Hence, y is a feasible point of (SLSVR). It also implies $F(y) = F(x)$ and that no upper level feasible point x' exists such that $F(x) \in F(x') + C_U \setminus \{0\}$.

Assume that y is not an efficient point of (SLSVR), meaning there exists a feasible point z such that

$$(F(y), \phi(F(y), y)) \in (F(z), \phi(F(z), z)) + C_{SLR} \setminus \{0\}. \quad (5)$$

This can only hold if $F(z) = F(y)$ and

$$\phi(F(y), y) \in \phi(F(z), z) + C_L \setminus \{0\}.$$

But then $(F(x), \phi(F(x), z))$ would dominate $(F(x), \phi(F(x), y))$ with respect to C_{SLR} . Consequently, (x, y) is not an efficient point of (HBP), which is a contradiction.

For the other direction, let y be an efficient point of (SLSVR). Clearly, the point (y, y) is a feasible point of (HBP). Assume that there exists another feasible point (x', y') such that

$$(F(x'), \phi(F(x'), y')) \leq_{C_{SLR} \setminus \{0\}} (F(y), \phi(F(y), y)).$$

This implies $F(x') \leq_{C_U} F(y)$. Due to the constraints, it follows that $F(y') \leq_{C_U} F(x')$. Together, this leads to $F(y') \leq_{C_U} F(x') \leq_{C_U} F(y)$.

Since y' is a feasible point of (SLSVR) for y to be efficient, the above inequality can only hold if $F(y') = F(x') = F(y)$. Consequently, $\phi(F(y), y)$ must be in $\phi(F(x'), y') + C_L \setminus \{0\}$ for the assumption to hold. Therefore, y cannot be an efficient point of (SLSVR), which is a contradiction. $\square$

To calculate an optimal point of an MCO problem a common method is to solve a corresponding weighted sum problem with weights from the dual cone of the domination cone. The obtained solution is weakly non-dominated (see Lemma 1). Unfortunately, due to the construction of C_{SLR} the set of weakly non-dominated points is far too large to be usable as solution of the optimization problem, since all points that are weakly non-dominated in the upper level are weakly non-dominated with regard to C_{SLR} , independent of their lower level value. To distinguish points that are the same in the upper level but different in the lower level is the conceptual idea behind the hierarchical bi-level problem. Hence, the weakly non-dominated solutions are meaningless. The points of interest are the non-dominated ones. However, as the interior of C_{SLR}^* is empty, there are no weights in C_{SLR}^* that guarantee true optimality with regard to C_{SLR} . Consequently, the cone is unfit for calculations. To overcome this limitation, we approximate C_{SLR} . The cone can be reformulated as

$$C_{SLR} = ((C_U \setminus \{0\}) \times \mathbb{R}^{n_L}) \cup (\{0\} \times C_L) \\ = (C_U \times \mathbb{R}^{n_L}) \setminus (\{0\} \times C'_L),$$

where C'_L denotes the complement of C_L in $\mathbb{R}^{n_L}$, that is, $C'_L = \mathbb{R}^{n_L} \setminus C_L$. If C_L and C_U are the standard domination cone, one obtains

$$C_{SLR}(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L}) = (\mathbb{R}_{\geq 0}^{n_U} \times \mathbb{R}_{\geq 0}^{n_L}) \setminus (\{0\} \times (\mathbb{R}_{\geq 0}^{n_L})').$$

To get more non-dominated points we need a smaller approximation cone. The idea of the approximation cone is to exclude slightly more than $(\{0\} \times (\mathbb{R}_{\geq 0}^{n_L})')$. Based on the standard domination cones for each level, we propose the following approximation.

Definition 2.11. Let Id^n denote the identity matrix of dimension $n \times n$ and $\mathbf{0}^{n,m}$ a matrix with only zeros of size $n \times m$. Then the approximation cone $C_A(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L})$ for $C_{SLR}(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L})$ is the set

$$C_A = \{Q\lambda \mid \lambda \geq 0\},$$

with

$$Q = \begin{pmatrix} Id^{n_U} & \mathbf{0}^{n_U, n_L} \\ \mathbf{0}^{n_L, n_U} & Id^{n_L} \end{pmatrix} A^1 \dots A^{n_L} \in \mathbb{R}^{n_U + n_L, n_U + n_L + n_U \cdot n_L},$$

where

$$A^k = \begin{pmatrix} \text{diag}(\epsilon_k) \\ \mathbf{0}^{k-1, n_U} \\ -1 \\ \mathbf{0}^{n_L-k, n_U} \end{pmatrix} \in \mathbb{R}^{n_U + n_L, n_U}, \text{ for } k = 1, \dots, n_L,$$

and $\epsilon_k > 0$ is a vector with n_U entries. Each entry $\epsilon_{k,j}$ for $k = 1, \dots, n_L$, $j = 1, \dots, n_U$ describes a minimal trade off such that a point $p = (p_u, p_l)^T \in \mathbb{R}^{n_U + n_L}$ with negative value for p_{l_k} can still be in the cone if there is one j such that $p_{u_j} \geq -\epsilon_{k,j} p_{l_k}$. All

parameters $\epsilon_{k,j}$ together will be denoted as matrix $\mathcal{E}$ and referred to as $\mathcal{E}$ -matrix.

We denote the dependence of C_A on the $\mathcal{E}$ -matrix as $C_{A(\mathcal{E})}$, whenever the choice of $\mathcal{E}$ is not clear. Notice how the blocks from Q stem from the upper and lower level domination cones, with the concept that a point where the lower level part is outside of C_L leads to costs in the upper level. As such the concept of the presented approximation can also be transferred to polyhedral domination cones C_U and C_L .

We will write λ as $(\lambda_u, \lambda_l, \lambda_{A^1}, \dots, \lambda_{A^{n_L}}) \in \mathbb{R}_{\geq 0}^{n_U} \times \mathbb{R}_{\geq 0}^{n_L} \times \mathbb{R}_{\geq 0}^{n_U} \times \dots \times \mathbb{R}_{\geq 0}^{n_U}$ to refer to the entries of λ that are multiplied with the respective blocks of Q . This means a point in C_A can be written as

$$\begin{pmatrix} \lambda_{u_1} + \sum_{k=1}^{n_L} \epsilon_{k1} \lambda_{A^k} \\ \vdots \\ \lambda_{u_{n_U}} + \sum_{k=1}^{n_L} \epsilon_{kn_U} \lambda_{A^k} \\ \lambda_{l_1} - \sum_{j=1}^{n_U} \lambda_{A^j} \\ \vdots \\ \lambda_{l_{n_L}} - \sum_{j=1}^{n_U} \lambda_{A^{n_L}} \end{pmatrix}.$$

C_A is a proper cone and its dual cone has non-empty interior.

Lemma 3. C_A^* has non-empty interior.

Proof. The dual cone can be expressed as:

$$C_A^* = \{y \in \mathbb{R}^{n_U + n_L} \mid Q^T y \geq 0\} \\ = \left\{ y \in \mathbb{R}^{n_U + n_L} \mid \begin{pmatrix} y_u \\ y_l \\ A^{1^T} y \\ \vdots \\ (A^{n_L})^T y \end{pmatrix} \geq 0 \right\}.$$

The first set of inequalities implies that $\mathbb{R}_{\geq 0}^{n_U} \times \mathbb{R}_{\geq 0}^{n_L} \supseteq C_A^*$. The inequalities of the second part, $A^{k^T} y \geq 0$, imply

$$A^{k^T} y = \underbrace{\text{diag}(\epsilon_k)}_{\geq 0} y_u - \begin{pmatrix} y_{l_k} \\ \vdots \\ y_{l_k} \end{pmatrix} \geq 0 \text{ for } k = 1, \dots, n_L. \quad (6)$$

By choosing $y_u \in \text{int}(\mathbb{R}_{\geq 0}^{n_U})$, the first term becomes positive and $ay_u \in \mathbb{R}_{\geq 0}^{n_U}$ for any $a \in \mathbb{R}_{\geq 0}$. Thus, for any $y_l \in \mathbb{R}_{\geq 0}^{n_L}$, there exists a $y_u \in \text{int}(\mathbb{R}_{\geq 0}^{n_U})$ and a fixed factor $a \in \mathbb{R}_{\geq 0}$ such that $(a \cdot y_u, y_l) \in C_A^*$.

Let $y \in C_A^*$ be a fixed point with $y_u \in \text{int}(\mathbb{R}_{\geq 0}^{n_U})$ and $y_l \in \text{int}(\mathbb{R}_{\geq 0}^{n_L})$. Let for $z := (ay_u, y_l)$ the factor a be chosen such that for some positive factor b it holds that $A^{k^T} y > b$ for all $k = 1, \dots, n_L$.

To demonstrate that $\text{int}(C_A^*) \neq \emptyset$, we show that there exists a $\hat{\delta}$ -ball around z for some $\hat{\delta} > 0$ that is in C_A^* . Choose some δ sufficiently small such that a δ -ball in $\mathbb{R}^{n_U}$ around ay_u is in $\mathbb{R}_{\geq 0}^{n_U}$ and a δ -ball in $\mathbb{R}^{n_L}$ around y_l is in $\mathbb{R}_{\geq 0}^{n_L}$. Now, we choose $\delta \in \mathbb{R}^{n_U + n_L}$ such that

$$\begin{aligned} -\tilde{\delta}_{l_k} r_{l_k} &\geq -\frac{1}{2}b \quad \forall r \in \mathbb{R}^{n_U+n_L} \text{ with } \|r\|_2=1, \text{ for } k=1, \dots, n_L \\ \tilde{\delta}_{u_k} r_{u_k} e_k &\geq -\frac{1}{2}b \quad \forall r \in \mathbb{R}^{n_U+n_L} \text{ with } \|r\|_2=1, \text{ for } k=1, \dots, n_U. \end{aligned}$$

Then, choose $\hat{\delta}$ as

$$\hat{\delta} = \min \left\{ \min_k \tilde{\delta}_{l_k}, \min_k \tilde{\delta}_{u_k}, \delta \right\}.$$

It follows that for an arbitrary direction r with $\|r\|_2 \leq 1$

$$\begin{aligned} A^{k^T}(z + \hat{\delta}r) &= \text{diag}(\epsilon_k)(ay_u + \hat{\delta}r_u) + \begin{pmatrix} -y_{l_k} + \hat{\delta}r_{l_k} \\ \vdots \\ -y_{l_k} + \hat{\delta}r_{l_k} \end{pmatrix} \\ &= \underbrace{\text{diag}(\epsilon_k)ay_u - \begin{pmatrix} y_{l_k} \\ \vdots \\ y_{l_k} \end{pmatrix}}_{=A^{k^T}y > b} + \underbrace{\hat{\delta}\text{diag}(\epsilon_k)r_u - \hat{\delta}\begin{pmatrix} r_{l_k} \\ \vdots \\ r_{l_k} \end{pmatrix}}_{\geq -\frac{1}{2}b} \\ &> 0. \end{aligned}$$

Hence, for all points in $B_{\hat{\delta}}(z)$, (6) also holds and thus the $\hat{\delta}$ -ball is in C_A^* . Consequently, $z \in \text{int}(C_A^*)$, which then cannot be empty. $\square$

As such C_A can be used as domination cone and weights from the interior of the dual cone will result in non- C_A -dominated points. The next step is to show that for $\mathcal{E} \rightarrow 0$ the front $P_{C_A(\mathcal{E})}$ converges towards $P_{C_{SLR}(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L})}$ in the sense that the Hausdorff

distance between the two sets converges to zero. For this we use the characterisation given by the following lemma to determine whether a point is in C_A .

Lemma 4. A point (u, l) is in $C_A(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L})$ if and only if there exists $\tau \in \mathbb{R}_{\geq 0}^{n_U+n_L}$ such that all of the following equations hold:

$$u_j \geq 0, \quad \forall j = 1, \dots, n_U; \quad (7)$$

$$\sum_{j=1}^{n_U} \tau_{j,i} = 1, \quad \forall i \text{ with } l_i < 0; \quad (8)$$

$$\sum_{i=1}^{n_L} (-\tau_{j,i} \epsilon_{ij} l_i) \leq u_j, \quad \forall j = 1, \dots, n_U; \quad (9)$$

$$\tau_{j,i} = 0, \quad \forall j = 1, \dots, n_U \text{ if } l_i \geq 0. \quad (10)$$

Proof. We start by demonstrating that for any point in $C_A(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L})$, there exists a $\tau \geq 0$ satisfying (7–10). By definition, $(u, l) \in C_A(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L})$ if there exists a $\lambda \geq 0$ such that

$$u_j = \lambda_{u_j} + \sum_{i=1}^{n_L} \epsilon_{ij} \lambda_{A_i^j}, \quad \forall j = 1, \dots, n_U \quad \text{and} \quad (11)$$

$$l_i = \lambda_{l_i} - \sum_{j=1}^{n_U} \lambda_{A_j^i}, \quad \forall i = 1, \dots, n_L. \quad (12)$$

Thus, (7) clearly holds. We choose

$$\tau_{j,i} = \begin{cases} \frac{\lambda_{A_j^i}}{\sum_{k=1}^{n_U} \lambda_{A_k^i}} & \forall i \text{ with } l_i < 0 \text{ and } \forall j = 1, \dots, n_U, \\ 0 & \text{otherwise.} \end{cases} \quad (13)$$

Since a negative l_i implies the existence of a $k \in \{1, \dots, n_U\}$ such that $\lambda_{A_k^i} > 0$, this expression is well-defined. Clearly, (8) and (10) hold. Denote the index sets as $I_- = \{i = 1, \dots, n_L | l_i < 0\}$ and $I_+ = \{i = 1, \dots, n_L | l_i \geq 0\}$. For all $i \in I_-$,

$$0 > l_i = \lambda_{l_i} + \sum_{k=1}^{n_U} -\lambda_{A_k^i} \Rightarrow \frac{\lambda_{l_i}}{\sum_{k=1}^{n_U} \lambda_{A_k^i}} < 1.$$

With the choice of τ according to (13), we obtain

$$\begin{aligned} \sum_{i=1}^{n_L} (-\tau_{j,i} \epsilon_{ij} l_i) &= -\sum_{i \in I_-} \epsilon_{ij} \frac{\lambda_{A_j^i}}{\sum_{k=1}^{n_U} \lambda_{A_k^i}} \left(\lambda_{l_i} - \sum_{k=1}^{n_U} \lambda_{A_k^i} \right) \\ &= \sum_{i \in I_-} \lambda_{A_j^i} \epsilon_{ij} - \epsilon_{ij} \lambda_{A_j^i} \underbrace{\frac{\lambda_{l_i}}{\sum_{k=1}^{n_U} \lambda_{A_k^i}}}_{0 \leq \dots < 1} \\ &= \sum_{i \in I_-} \epsilon_{ij} \lambda_{A_j^i} \underbrace{\left(1 - \frac{\lambda_{l_i}}{\sum_{k=1}^{n_U} \lambda_{A_k^i}} \right)}_{0 < \dots \leq 1} \\ &\leq \sum_{i \in I_-} \epsilon_{ij} \lambda_{A_j^i} \\ &\leq u_j. \end{aligned}$$

Thus, the condition (9) is fulfilled.

The other direction is to show that the existence of a τ as described in (7–10) implies the existence of a $\lambda \geq 0$ for (11) and (12). Because the columns of Q span the whole $\mathbb{R}^{n_U+n_L}$, there is always a solution to $Q\lambda = (u, l)$. It remains to determine that one such solution is non-negative. Choose

$$\lambda_{A_j^i} = \begin{cases} -\tau_{j,i} l_i & \text{if } i \in I_-, \\ 0 & \text{if } i \in I_+. \end{cases} \quad (14)$$

Then for $i \in I_+$, it follows that $\lambda_{l_i} = l_i \geq 0$. For $i \in I_-$,

$$\begin{aligned} l_i &= \lambda_{l_i} - \sum_{j=1}^{n_U} \lambda_{A_j^i} \Leftrightarrow l_i + \sum_{j=1}^{n_U} \lambda_{A_j^i} \\ &= \lambda_{l_i} \Leftrightarrow l_i - \sum_{j=1}^{n_U} (\tau_{j,i} l_i) = \lambda_{l_i} \stackrel{(8)}{\Leftrightarrow} \lambda_{l_i} = 0. \end{aligned}$$

Concerning λ_u , we obtain

$$\begin{aligned} \sum_{i \in I_-} (-\tau_{j,i} \epsilon_{ij} l_i) &\stackrel{(10)}{=} \sum_{i=1}^{n_L} (-\tau_{j,i} \epsilon_{ij} l_i) \stackrel{(9)}{\leq} u_j \\ &\stackrel{(11)}{=} \lambda_{u_j} + \sum_{i=1}^{n_L} \epsilon_{ij} \lambda_{A_i^j} \stackrel{(14)}{=} \lambda_{u_j} - \sum_{i \in I_-} \epsilon_{ij} \tau_{j,i} l_i. \end{aligned}$$

Thus, the existence of a feasible $\lambda_u \geq 0$ is guaranteed. $\square$

For the convergence proof, we further make use of the following lemma about the closedness of non-dominated sets. For ease of notation we denote $P_{C_{A(\mathcal{E}_n)}}$ as P_n .

Lemma 5. *If $\mathcal{E}_n$ and $\mathcal{E}_m$ are two $\mathcal{E}$ -matrices with $\mathcal{E}_n \leq \mathcal{E}_m$ and P_m is closed, then P_n is also closed.*

Proof. With $C_{A(\mathcal{E}_n)} \supseteq C_{A(\mathcal{E}_m)}$ it directly follows that $P_n \subseteq P_m$. Define

$$X_0 := P_m + C_{A(\mathcal{E}_m)}$$

and

$$X := P_n + C_{A(\mathcal{E}_n)}.$$

Then X_0 is closed and X can also be expressed as $X = P_m + C_{A(\mathcal{E}_n)}$ and is thus also closed. Hence, the boundaries of X_0 and X are closed. Let

$$M := \text{bnd}(X_0) \cap \text{bnd}(X),$$

which is consequently also closed. Then

$$P_n = M \cap P_m$$

is as the intersection of two closed sets also closed. $\square$

Theorem 2.3. *Let $(\mathcal{E}_n)_{n=1}^\infty$ denote a sequence of $\mathcal{E}$ -matrices with entries ϵ_{ij} , such that $\mathcal{E}_n \geq \mathcal{E}_{n+1}$ component wise and $\mathcal{E}_n \rightarrow 0$ for $n \rightarrow \infty$. If for an $m \in \mathbb{N}$ the non- $C_{A(\mathcal{E}_m)}$ -dominated set P_m is closed and the objectives are bounded on the corresponding efficient set, then*

$$P_{C_{A(\mathcal{E}_n)}} \xrightarrow{n \rightarrow \infty} P_{C_{SLR}}(\mathbb{R}_{\geq 0}^{n_U}, \mathbb{R}_{\geq 0}^{n_L}).$$

Proof. For a sequence of $\mathcal{E}$ -matrices $(\mathcal{E}_n)_{n=1}^\infty$ with $\mathcal{E}_n \geq \mathcal{E}_{n+1}$ it follows that $C_{A(\mathcal{E}_n)} \subseteq C_{A(\mathcal{E}_{n+1})}$, yielding $P_n \supseteq P_{n+1}$. Note that $(P_n)_{n=1}^\infty$ forms a monotonically non-increasing sequence of sets in $\mathbb{R}^{n_U+n_L}$, and is compact for $n \geq m$, thus converging to a limit. We must show that the limit is $P_{C_{SLR}}$.

Let $\kappa > 0$. We demonstrate that there exists an n such that for all $\tilde{n} > n$ the Hausdorff distance between $P_{\tilde{n}}$ and $P_{C_{SLR}}$ is less than κ .

Let y be a point in P_m but not in $P_{C_{SLR}}$ and let x be a point in $P_{C_{SLR}}$ that dominates y with respect to C_{SLR} such that $\|x - y\| > \kappa$, where the norm denotes an arbitrary, fixed p -norm. Let $z := y - x$, and denote the variables belonging to the upper or lower level as z_u and z_l , respectively, so that $z = (z_u, z_l)$.

By assumption, $z \in C_{SLR}$ and $z \notin C_{A(\mathcal{E}_m)}$. Because $z \in C_{SLR}$, it follows that $z_u \geq 0$. If $z_u = 0$, then $z_l \in \mathbb{R}_{\geq 0}^{n_L}$. This would imply

$z \in C_{A(\mathcal{E}_m)}$. Consequently, there exists a $j \in \{1, \dots, n_U\}$ such that $z_{u_j} > 0$. Furthermore, if $z_l \in \mathbb{R}_{\geq 0}^{n_L}$, it also follows that $z \in C_{A(\mathcal{E}_m)}$. Hence, there is an $i \in \{1, \dots, n_L\}$ such that $z_{l_i} < 0$. Let $I_- := \{i = 1, \dots, n_L \mid z_{l_i} < 0\}$ and $I_+ := \{i = 1, \dots, n_L \mid z_{l_i} \geq 0\}$.

For the first case, we assume there is a $j \in \{1, \dots, n_U\}$ with $z_{u_j} > \kappa$. By assumption, $|z_{l_i}|$ is bounded by a threshold t_i . For any $z_{u_j} > \kappa$ and $i \in I_-$, we define $\epsilon_{ij} := \frac{\kappa}{t_i n_L} > 0$. Then,

$$\begin{aligned} \sum_{i=1}^{n_L} (-\tau_{j,i} \epsilon_{ij} z_{l_i}) &\leq \sum_{i \in I_-} (-\tau_{j,i} \epsilon_{ij} z_{l_i}) = \sum_{i \in I_-} \tau_{j,i} \frac{\kappa}{t_i n_L} |z_{l_i}| \\ &\leq \sum_{i \in I_-} \tau_{j,i} \frac{\kappa}{t_i n_L} t_i = \frac{\kappa}{n_L} \sum_{i \in I_-} \tau_{j,i}. \end{aligned}$$

For any τ with $\sum_{i \in I_-} \tau_{j,i} \leq 1$, it follows that

$$\frac{\kappa}{n_L} \sum_{i \in I_-} \tau_{j,i} \leq \kappa < z_{u_j}, \quad (15)$$

and consequently, the inequality of (9) holds for the j chosen above. To find a feasible τ , the second case, where $z_{u_k} < \kappa$ for all $k \in \{1, \dots, n_U\}$, also has to be covered. Hence, we consider one such arbitrary, fixed $k \in \{1, \dots, n_U\}$ with $z_{u_k} > 0$. Define $y' := (y_{u_1}, \dots, y_{u_{k-1}}, x_{u_k}, y_{u_{k+1}}, \dots, y_{u_{n_U}}, y_{l_1}, \dots, y_{l_{n_L}})$. This point y' cannot belong to $P_{C_{SLR}}$ or P_m , as y' C_A -dominates y , contradicting our assumption. Since P_m is closed, there exists a $\delta > 0$ such that the δ -ball $B_\delta(y')$ around y' has no common point with P_m , particularly $y \notin B_\delta(y')$. Thus, $\|y' - y\|_\infty \geq \delta$, which implies $z_{u_k} \geq \delta$. Defining $\epsilon_{i,k} = \frac{\delta}{t_i n_L} > 0$ leads to an analogous estimation as in the first case.

If there exists a τ satisfying Lemma 4, then z is in $C_{A(\mathcal{E}_n)}$ with the corresponding $\mathcal{E}_n$ based on the chosen ϵ_{ij} . The existence of such a τ is trivial. We fix an arbitrary j with $z_{u_j} > 0$ from either case and choose

$$\tau_{k,i} = \begin{cases} 1 & \text{for } i \in I_-, k=j \\ 0 & \text{otherwise.} \end{cases}$$

Then clearly $\sum_{i \in I_-} \tau_{j,i} \leq |I_-| \leq n_L$ and thus the estimation in (15) holds. Moreover, with $\tau_{k,i} = 0$ for $k \neq j$, this means (9) holds. That Equations (8) and (10) hold is a direct consequence of the choice of τ . Therefore, with Lemma 4 we conclude that z is in $C_{A(\mathcal{E}_n)}$ with ϵ_{ij} as defined above.

Consequently, $P_{\hat{n}} \rightarrow P_{C_{SLR}}$ as $\hat{n} \rightarrow \infty$ with respect to the maximum norm as distance measure for the Hausdorff metric. Since the maximum norm on $\mathbb{R}^{n_U+n_L}$ is equivalent to all other p norms, we also have convergence with respect to any of them. $\square$

While the weakly non-dominated points with respect to C_A still include all points that are weakly non-dominated in the upper level independent of the lower level, the interior of the dual cone C_A^* is not empty. Thus, contrary to C_{SLR} it is possible to calculate truly non-dominated points with C_A . In the next

section, we introduce the reduction approach as one of two methods to calculate the non-dominated points of HBP. The reduction approach makes use of all theoretic findings presented in this section.

2.3 | Solving the Hierarchical Bi-Level Problem

In this section, we introduce two different approaches to solve the multi-criteria hierarchical bi-level problem (HBP). The performance of these two methods will later be compared in the results section when applied to specific proton treatment planning examples. As a subroutine, both methods use the Sandwiching algorithm described in Serna (2012) and Bokrantz and Forsgren (2012). The pseudo code for the sandwiching algorithm can be found in Bokrantz and Forsgren (2012, Algorithm 3.1). For our purposes, we define a stopping criterion for the Sandwiching algorithm that is met if either the approximation error reaches a given target value δ , or the number of computed solutions reaches a given maximum number N . We employ the following signature

$$X^* = \text{Sandwiching}(\mathcal{M}, \delta, N, C)$$

where X^* is the set of calculated Pareto efficient solutions and $\mathcal{M}$ is the problem instance. C denotes the domination cone, which can be any proper cone. In the standard case, C is the positive orthant $\mathbb{R}_{\geq 0}^n$.

2.3.1 | Two-Step Approach

The first algorithm can be seen as a generalisation of the approach of Unkelbach et al. (2016). It operates in two stages. In the *first stage*, a set of Pareto efficient solutions $X_u = x_u^1, \dots, x_u^N$ to the (multi-criteria) upper level problem

$$\begin{aligned} \min_{x_u} \quad & F(x_u) \\ \text{s. t.} \quad & G_i(x_u) \leq 0, i = 1, \dots, I. \end{aligned} \quad (16)$$

is calculated using the Sandwiching algorithm. In the *second stage*, for each solution x_u^* from the first stage, the Pareto front of the problem

$$\begin{aligned} \min_{x_l} \quad & \phi(x_l) \\ \text{s. t.} \quad & \psi_j(x_l, F(x_u^*)) \leq 0, j = 1, \dots, J \end{aligned} \quad (17)$$

is computed, again using the Sandwiching algorithm. Problem (17) is the lower level problem of (HBP), where the upper level variable x_u is fixed to x_u^* . In Algorithm 1 the pseudo code for the two-step approach is depicted.

2.3.2 | Reduction Approach

The theoretical findings of Section 2.2 motivate a second, potentially more efficient method for solving (HBP). As shown in Theorem 2.1, it is possible to reformulate the multi-criteria hierarchical bi-level problem (HBP) to a single-level

multi-criteria problem (MCO) by defining a specific domination cone. Definition 2.11 describes a polyhedral cone C_A that approximates $C_{SLR}(\mathbb{R}_{\geq 0}^{n_u}, \mathbb{R}_{\geq 0}^{n_l})$ and whose dual has non-empty interior. As the Sandwiching algorithm can be applied using any polyhedral cone (Serna 2012; Bokrantz and Forsgren 2012), this allows the direct application of the Sandwiching algorithm to the reformulated problem with the approximation cone C_A . Because of the convergence result from Theorem 2.3, we can expect to obtain a set of solutions which approximates the front of HBP reasonably well provided that the entries of $\mathcal{E}$ are chosen suitably small. We call this the *reduction approach*. In Algorithm 2 the pseudo code for the reduction approach is depicted.

3 | Results

In this section, we apply the two algorithms from Section 2.3—the two-step approach and the reduction approach—to calculate the fronts for two realistic proton treatment planning problems. The problems were taken from the CORT data set (Craft et al. 2014), and their dose and LET mappings were calculated with *matRad* (Cisternas et al. 2015), an open-source software for radiation treatment developed by the German Cancer Research Center. For both approaches, we employ the commercially available nonlinear optimization solver *knitro* (Byrd et al. 2006) to solve the weighted sum problems within the Sandwiching subroutine.

As motivated in the introduction, we conceptualise the treatment planning problem as a multi-criteria hierarchical bi-level problem (HBP). For the reoptimization to include LD objectives, we set the allowed deviation from the upper level (dose objectives) to zero. This enabled us to apply the SLSVR approach. Consequently, the dimension of the decision space of the reduced problem is equal to the dimension of the decision space of the initial dose planning problem. Each upper level objective F_i , each lower level objective ϕ_j , and each constraint defining the feasible set X —in addition to the physical constraint of positive irradiation—employs one of the following

ALGORITHM 1 | Two-step approach.

input:

- a multi-criteria hierarchical bi-level problem (HBP)
- a target approximation quality $\delta_1 \geq 0$ and a maximal number of solutions $N_1 \geq 0$ (for first stage)
- a target approximation quality $\delta_2 \geq 0$ and a maximal number of solutions $N_2 \geq 0$ (for second stage)

output:

- a set of efficient solutions X^* to (HBP)

Start

$X_u = x_u^1, \dots, x_u^N \leftarrow \text{Sandwiching}((16), \delta_1, N_1)$

for all $x_u^k \in X_u$ **do**

$X^{*,k} \leftarrow \text{Sandwiching}((17), \delta_2, N_2)$

end for

return $\cup_{k=1}^N X^{*,k}$

End

evaluation functions to evaluate the dose distribution (physical or LETxD) in a given volume V , which can be a target or risk structure.

The *underdose* penalises any voxel dose below the reference value d^{ref} :

$$\text{UD}(d) = \sum_{v \in V} (\max \{0, d^{\text{ref}} - d_v\})^2.$$

The *overdose* penalizes any voxel dose above the reference value

$$\text{OD}(d) = \sum_{v \in V} (\max \{0, d_v - d^{\text{ref}}\})^2.$$

The *generalised equivalent uniform dose* for an exponent $p \geq 1$ is

$$\text{gEUD}(d) = \sum_{v \in V} d_v^p.$$

ALGORITHM 2 | Reduction approach.

input:

- a multi-criteria hierarchical bi-level problem (HBP)
- a target approximation quality $\delta \geq 0$ and a maximal number of solutions $N \geq 0$
- an $\epsilon > 0$ to define the domination cone

output:

- a set of feasible solutions X^* to (HBP) whose images approximate the front of (HBP)

Start

Reformulate (HBP) to (MCO) according to Theorem 2.1

Define the domination cone C_A of 2.11 with $\epsilon_{ij} := \epsilon$

return *Sandwiching* $((\text{MCO}), \delta, N, C_A)$

End

To calculate the SLSVR we used an approximated cone as introduced in Definition 2.11. Per computation, we chose one value for all ϵ_{ij} , which we will henceforth refer to as ϵ . To observe the effects of the approximation, we varied the value of ϵ .

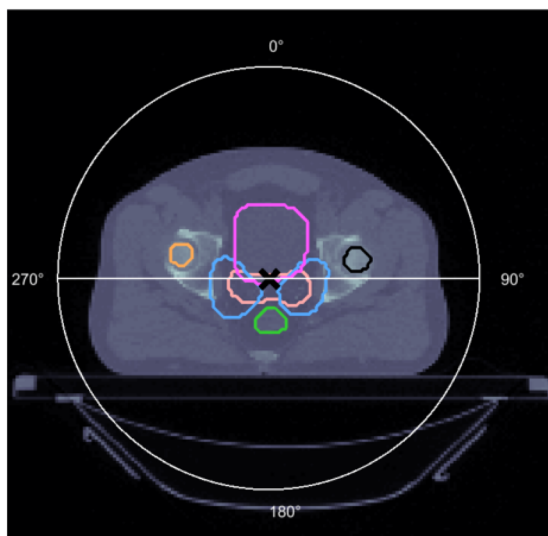
3.1 | Prostate Case

Our first example was a prostate case. The target was irradiated from two proton beams from opposing sides (90° and 270°). There is a primary tumour (PTV 68) with a prescription dose of 68 Gy and a larger surrounding target (PTV 56) with a prescription dose of 56 Gy. The relevant risk structures were the rectum, the bladder, and the left and right femoral heads; see Figure 1a.

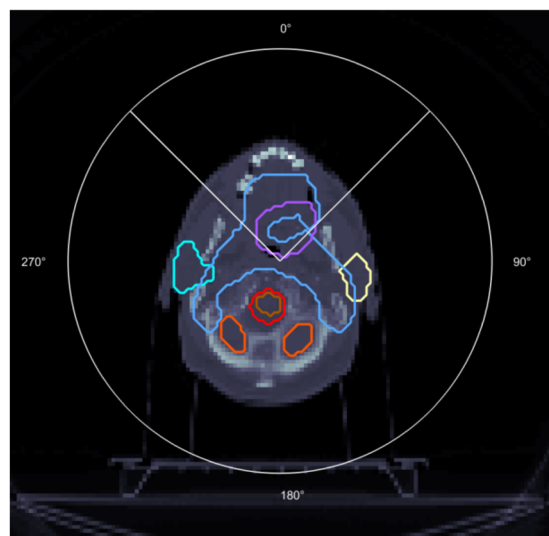
We investigated an optimization model with two upper-level objectives measuring the physical dose in the bladder and rectum, and two lower-level objectives measuring the LETxD in the bladder and rectum. Target coverage, the sparing of the femoral heads and the unclassified tissue (body) was ensured by constraints. The optimization model is depicted in Table 1.

We ran the two-step approach with $N_1 = 10$, $N_2 = 5$ and the reduction approach with $N = 50$. We also set all target approximation qualities to 0, such that for both approaches, the output set consisted of exactly 50 solutions. For the approximating cone in the reduction approach, we employed $\epsilon = 0.01$, $\epsilon = 0.025$, $\epsilon = 0.05$, and $\epsilon = 0.1$.

Figure 2 shows the objective space images of the calculated solution sets. Figure 2a,c,e,g are the projections on the upper level objectives evaluating the physical dose, while Figure 2b,d,f,h are the projections on the lower level objectives evaluating LETxD. Recall that the two-step approach guarantees Pareto optimality with respect to the upper level objectives, while the points



(a) Prostate case: PTV 56 (rose), PTV 68 (blue), bladder (pink), rectum (green), left femoral head (black), right femoral head (orange).



(b) Head and neck case: PTV 70 (pink), PTV 63 (blue), right parotid (light blue), left parotid (yellow), cerebellum (orange), brain stem (brown), spinal cord (red).

FIGURE 1 | Geometry of both cases. Images were created with *MatRad* (Cisternas et al. 2015).

obtained from the reduction approach only create an approximation that improves with decreasing ϵ . This is confirmed in the left hand side of Figure 2, showing the upper level (dose) objective evaluations to be worse for the reduction approach. However, the difference between the point sets of the two approaches gets much smaller as ϵ decreases.

On the other hand, the points obtained from the reduction approach reach better values for the lower level (LETxD) objectives, as can be seen in the right column of Figure 2. The reason for this is that for the reduction approach, but not the two-step approach, some relative trade-off between upper and lower level objectives is allowed. Again, the difference between the set of points decreases as ϵ decreases.

To further quantify this close approximation of the two-step front by the SLSVR front with decreasing ϵ , we also calculated the Hausdorff distance between the images of both representative sets. To place less emphasis on how well the methods represent different areas of the front, only points that were in the range covered by the two-step approach were considered. Figure 3 shows the resulting distances for the different values of ϵ .

Figure 4 shows the dose volume histogram and the dose averaged LET distribution for two comparable plans marked in Figure 2g,h, one originating from the two-step approach and one from the reduction approach. Figure 5a–d shows the difference in the dose, LETxD and LET distribution in one plane for these selected plans from the two calculation approaches. Both figures demonstrate a small dose difference, but significantly reduced LET for the reduction plan compared to the two-step plan.

TABLE 1 | The optimization problem for the prostate case.

Volume	Evaluation function	Dose type
Lower level objectives		
Rectum	$\text{gEUD}, p = 2$	LETxD
Bladder	$\text{gEUD}, p = 2$	LETxD
Upper level objectives		
Rectum	$\text{gEUD}, p = 2$	Physical
Bladder	$\text{gEUD}, p = 2$	Physical
Upper level constraints		
PTV 68	$\text{UD}, d^{\text{ref}} = 68, \leq 10$	Physical
PTV 68	$\text{OD}, d^{\text{ref}} = 73, \leq 5$	Physical
PTV 56	$\text{UD}, d^{\text{ref}} = 56, \leq 10$	Physical
PTV 56	$\text{OD}, d^{\text{ref}} = 61, \leq 5$	Physical
Rt femoral head	$\text{gEUD}, p = 2, \leq 250$	Physical
Lt femoral head	$\text{gEUD}, p = 2, \leq 250$	Physical
Body	$\text{OD}, d^{\text{ref}} = 30, \leq 10$	Physical

3.2 | Head and Neck Case

As a second example, we looked at a head and neck case with two beams from 45° to 315° respectively. There is a primary tumour (PTV 70) with a prescription dose of 70 Gy and a larger surrounding target (PTV 63) with a prescription dose of 63 Gy. The relevant risk structures are the left and right parotid, brain stem, cerebellum, spinal cord and larynx; see Figure 1b.

Again, we investigated an optimization model with two upper level objectives, this time measuring the physical dose in the left and right parotid, and two lower level objectives, measuring LETxD in the left and right parotid. Target coverage and the sparing of brain stem, spinal cord, cerebellum and skin is ensured by constraints. The optimization model is given in Table 2.

We ran the two-step approach with $N_1 = N_2 = 5$ and the reduction approach with $N = 25$ to obtain exactly 25 solutions with both approaches. For the approximating cone in the reduction approach, we employed both $\epsilon = 0.1$ and $\epsilon = 0.01$.

Figure 6 shows the objective space images of the calculated solution sets as projections on the upper and lower level objectives. As in the prostate example, and corresponding to theory, the upper level objective evaluations are worse for the reduction approach while the lower level objective evaluations are better. As expected, the difference between the point sets shrinks significantly when switching from $\epsilon = 0.1$ to $\epsilon = 0.01$.

Figure 7 shows the dose volume histogram and the dose averaged LET distribution for two comparable plans obtained from the two different approaches, marked in cyan (SLSVR) and orange (two-step) in Figure 6a,b. Figure 5e–h shows the difference in dose, LET and LETxD distribution for those two selected plans.

3.3 | Calculation Times Comparison

To compare the run times of both algorithms, we calculated a two-step plan for both cases with $N_1 = N_2 = 5$ and SLSVR plans with 25 points and with ϵ values 0.2, 0.1, and 0.05. Again, the approximation quality in the Sandwiching subroutine was set to 0, resulting in exactly 25 plans per calculation. All computations were performed on a Lenovo T490s laptop with a 13th Gen Intel Core i7-1370P 1.90 GHz processor. We observed a speed up when using the SLSVR algorithm of at least factor 3 in comparison to the two-step approach. The precise run times are listed in Table 3.

One reason for the longer calculation time of the two-step approach is that in the first stage, 5 additional plans have to be optimized. Also, the individual optimization runs took longer on average for the two-step approach, as more solver iterations were needed. One likely reason for this is that the tight constraints imposed on the second stage optimization problems make them more challenging to solve numerically. The calculation overhead on top of the individual optimization runs is negligible for both

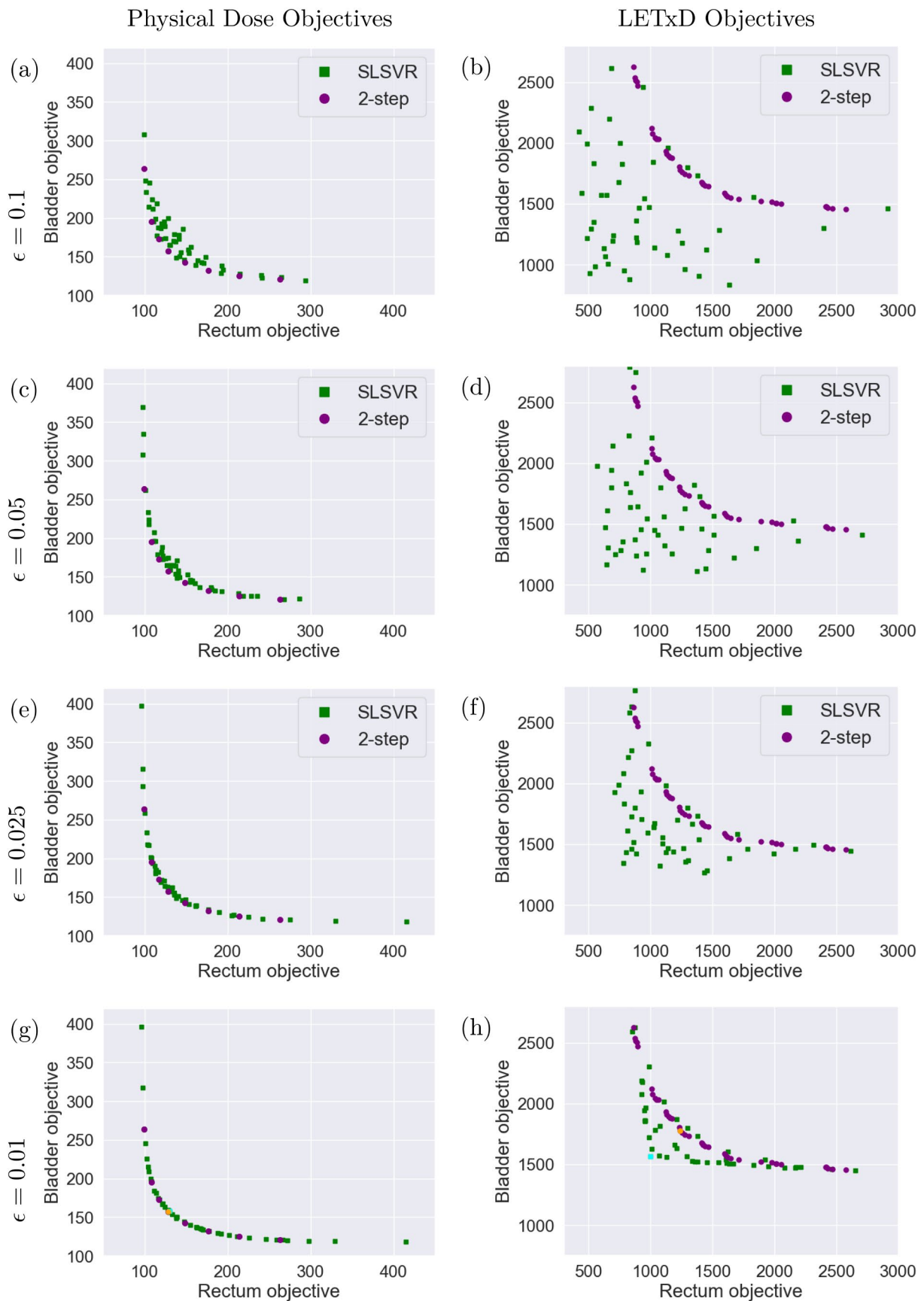


FIGURE 2 | Projections of the solutions from the two-step approach's second step (purple circles) and the SLSVR approach (green squares) for the prostate case. Left: Projection of the physical dose objectives. Right: Projection of the LETxD objectives. Each row is obtained with a different ϵ for the approximation cone $C_{A(\epsilon)}$. The points from the two-step approach are the same in all plots. The orange (2-step) and cyan (SLSVR) points in the last row are the plans used for further comparative analyses.

approaches and creates no meaningful difference in calculation time.

4 | Discussion

In this work, we described a concise and novel way of including LETxD in the plan optimization for proton treatment planning. To this aim, we defined a particular type of bi-level optimization problem which we called a hierarchical bi-level problem. We showed how this type of problem can be reduced to a single-level multi-criteria problem, and how this reduction can be utilized to calculate a representative set of the non-dominated points. This representation can serve as the plan set for a cost-based interactive multi-criteria proton treatment plan optimization approach.

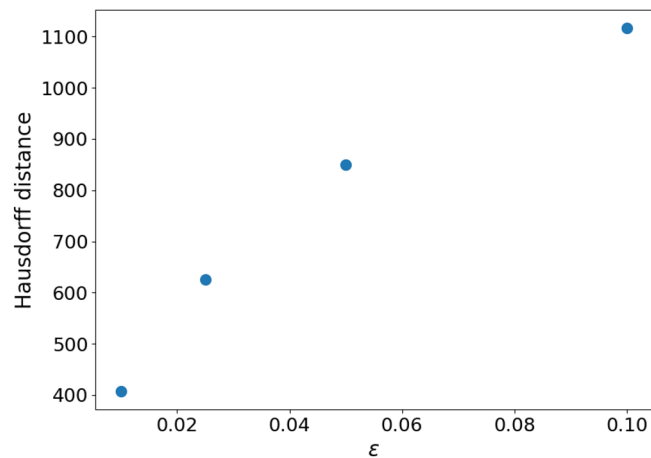


FIGURE 3 | Hausdorff distance between the images of the representative points obtained from the two-step and the SLSVR algorithm.

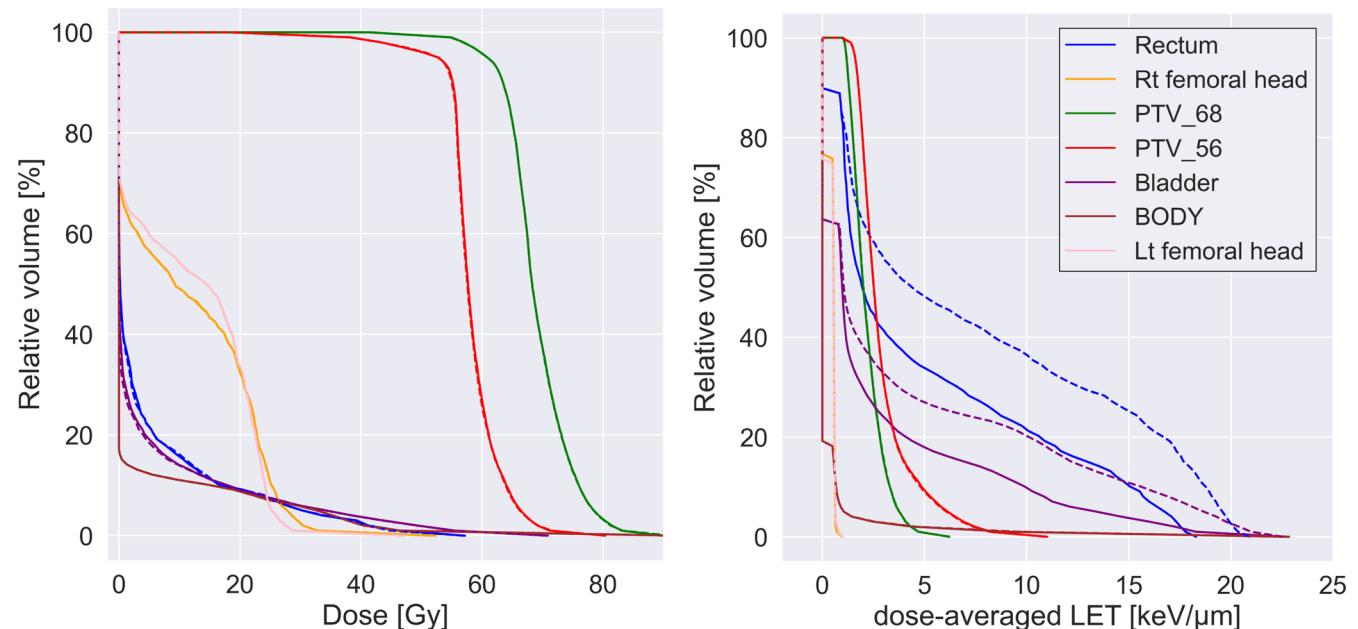


FIGURE 4 | Comparison of two similar optimized plans for the prostate case, one obtained from the two-step approach (highlighted in orange in Figure 2, shown as dashed line here), one from the reduction approach (highlighted in cyan in Figure 2). Left: Comparison of dose volume histograms showing, on the y-axis, the relative volume of a structure that receives at least the amount of dose on the x-axis. Right: Dose-averaged LET volume histogram, showing the relative volume of a structure that exhibits a dose averaged LET of at least the value on the x-axis.

The reduction approach has the advantage that algorithms from MCO can be utilized, leading to an algorithmic-specific good coverage of the front. Moreover, it allows controlling the relative trade-off between the two levels. We successfully demonstrated the application of the algorithm for two exemplary proton cases.

For hierarchical bi-level problems with convex objectives and constraints, the most significant advantage of the reduction approach lies in the direct application of the Sandwiching algorithm. By its use of inner and outer approximation, the Sandwiching algorithm is uniquely effective in creating a good representation of the front with very few weighted sum optimizations. In this paper, we did not investigate how effectively the calculated points represent the true front, as this would have been a computationally intractable task for the rather large exemplary problems we discussed. However, as a future research topic, a computational study on a set of benchmark problems with analytically characterized fronts could provide this insight.

The proofs in this paper show that for decreasing ϵ we get a better approximation of the efficient points. However, from a numerical perspective small values for ϵ also require stricter stopping criteria when solving the weighted-sum problems, as very small absolute values for the weights occur. In practice, this may limit the achievable approximation quality.

The presented reduction approach likewise works for non-convex hierarchical bi-level problems. To find a representation of the front, an algorithm that can handle non-convex problems has to be employed for the resulting single-level reduction. Sandwiching, based on the weighted sum, cannot find all non-dominated points of a non-convex problem. However, for example hyperboxing algorithms (e.g.,

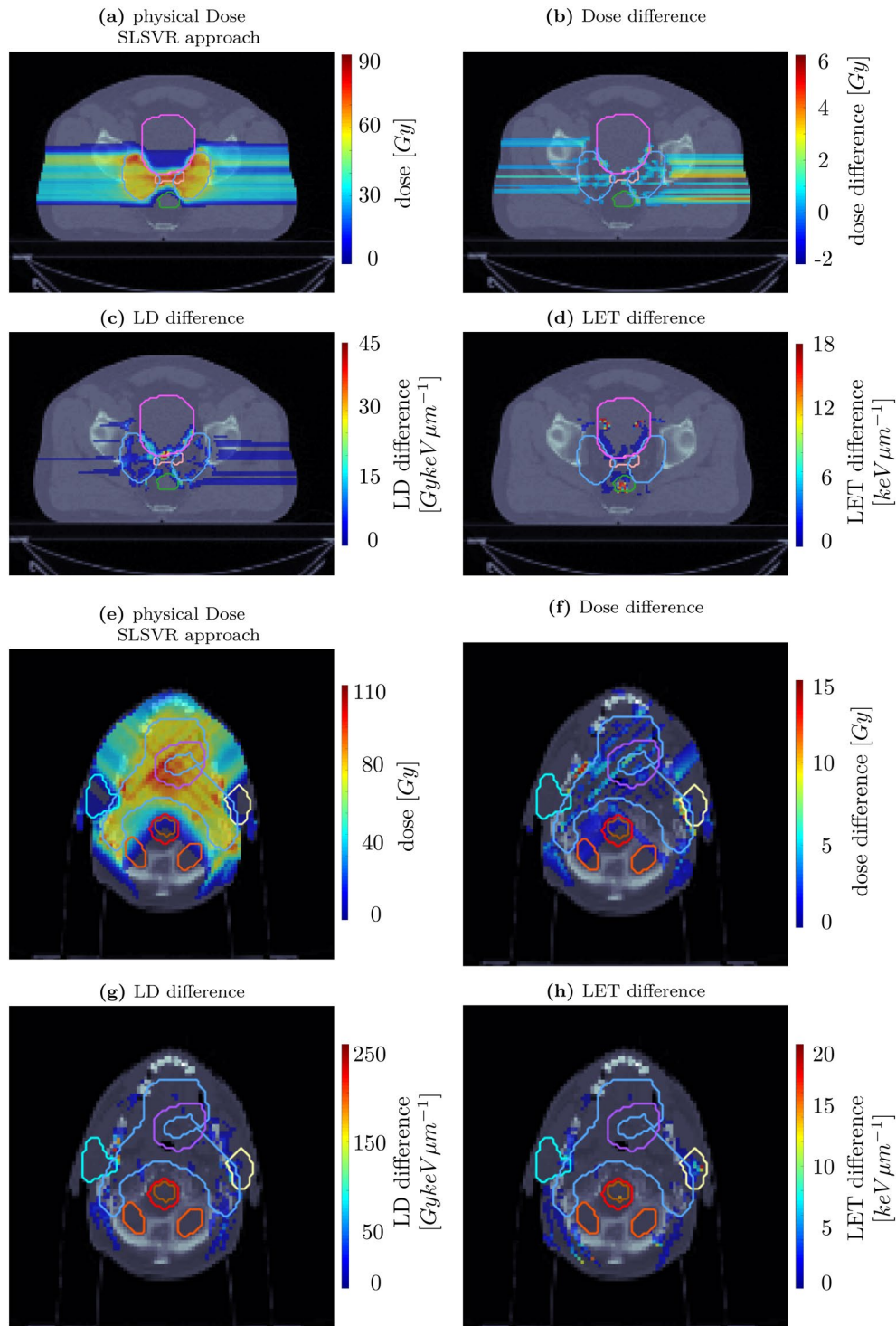


FIGURE 5 | Heat maps for the prostate case (a–d) and the head and neck case (e–h) showing: (a, e) the dose distribution of the selected plan from the SLSVR approach; (b–d, f–h) the differences of the two selected plans per case. A positive difference means the according value in the plan from the two-step approach is higher in comparison to the SLSVR approach.

Serna 2012; Teichert 2014; Klamroth et al. 2002; Eichfelder and Warnow 2023) can be used for this. To use a hyperboxing algorithm with the SLR approach, it has to be able to handle polyhedral domination cones. Such a generalisation is shown in Schubert (2024).

In the context of a cost-based interactive MCO setting, the generated plan database is subsequently used in the interactive, real-time decision-making process called *navigation* (Allmendinger et al. 2017). In current clinically used software for radiotherapy planning, navigation is facilitated by a collection of objective-associated

TABLE 2 | The optimization problem for the head and neck case.

Volume	Evaluation function	Dose type
Lower level objectives		
Left parotid	$\text{gEUD}, p = 2$	LETxD
Right parotid	$\text{gEUD}, p = 2$	LETxD
Upper level objectives		
Left parotid	$\text{gEUD}, p = 2$	Physical
Right parotid	$\text{gEUD}, p = 2$	Physical
Upper level constraints		
PTV 70	$\text{UD}, d^{\text{ref}} = 2.33, \leq 0.02$	Physical
PTV 70	$\text{OD}, d^{\text{ref}} = 2.56, \leq 0.01$	Physical
PTV 63	$\text{UD}, d^{\text{ref}} = 2.1, \leq 0.04$	Physical
PTV 63	$\text{OD}, d^{\text{ref}} = 2.3, \leq 0.01$	Physical
Skin	$\text{OD}, d^{\text{ref}} = 1, \leq 1.0$	Physical
Skin	$\text{OD}, d^{\text{ref}} = 2.67, \leq 0.01$	Physical
Skin	$\text{OD}, d^{\text{ref}} = 3.00, \leq 0.0$	Physical
Brain stem	$\text{gEUD}, p = 2, \leq 1.0$	Physical
Cerebellum	$\text{gEUD}, p = 2, \leq 2.0$	Physical
Spinal cord	$\text{gEUD}, p = 2, \leq 2.0$	Physical
Larynx	$\text{gEUD}, p = 2, \leq 1.0$	Physical
Spinal cord	$\text{gEUD}, p = 2, \leq 2.0$	Physical
Left parotid	$\text{gEUD}, p = 2, \leq 0.5$	Physical
Right parotid	$\text{gEUD}, p = 2, \leq 0.5$	Physical

Note: Model constraints were formulated in terms of fraction dose (30 fractions total).

sliders (Monz et al. 2008). For this method, the additional sliders associated with the LET-weighted dose objectives are not prohibitive from a computational perspective. However, the increase in cognitive load on the DM may be significant, which may nonetheless warrant a customised navigation scheme. The conception of such a scheme is a topic for further research.

For our numerical analysis, we investigated problems with two objectives each for both levels of the bi-level problem. In a clinical setting, the number of relevant structures can be higher, up to 5–8 depending on the tumour site. With increasing dimensionality of the problem, the numerical effort of our proposed reduction method will rise according to the underlying approximation scheme. For Sandwiching, it has been shown that the number of iterations needed to achieve a certain measure of approximation quality $\delta < 1$ increases in order $\mathcal{O}(\delta^{1-n})$ with the number of objective dimensions n . Under certain regularity conditions with respect to the differentiability and optimality conditions of the MCO problem, the number of iterations needed is in the order of $\mathcal{O}(\delta^{\frac{1-n}{2}})$ (Lammel 2024). In addition to the increase in the weighted sum problems to be solved, the internal representation of the Sandwiching approximation will also increase sharply with dimension (Bokrantz and Forsgren 2012). With these limitations, Bokrantz and

Forsgren (2012) conclude that a Pareto front approximation for up to 10 objectives is reasonable in a clinical setting with limited planning time. Hence, when using the reduction approach, the planner may need to proactively manage the problem size by including only the most important metrics as objectives, while reformulating other metrics as a constraint instead. However, compared to other bi-level reduction approaches as mentioned in the introduction, the dimension of the objective space of our approach is far lower.

Further, we observe that the reduction approach can be extended to multi-level problems. Lemma 2 showed that the domination cone C_{SLR} needed for the single-level reduction retains many properties from the domination cone C_U of the upper level and the domination cone C_L of the lower level. This means that C_{SLR} has all the required properties to function as C_L for the next higher level. Hence, multi-level problems can recursively be reduced to a single-level problem with the presented approach. For this, repeatedly the two inner most levels are reduced and then form the new inner most level.

Besides proton therapy planning, hierarchical bi-level problems can be used to model a wide range of application scenarios in which a solution is first optimized according to certain criteria and later re-optimized according to certain other criteria.

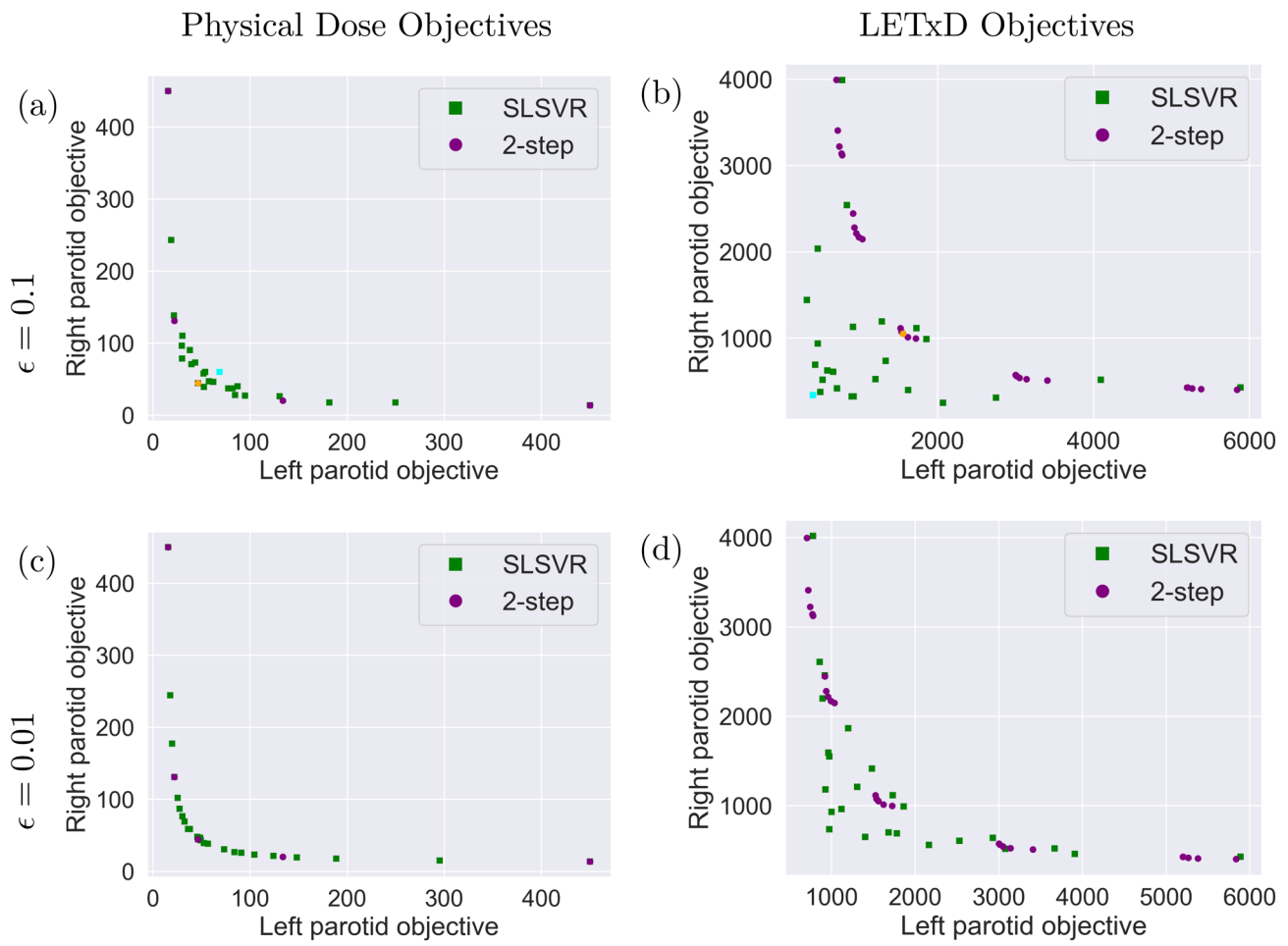


FIGURE 6 | Projections of the solutions from the two-step approach's second step (purple circles) and the SLSVR approach (green squares) for the head neck case. Left: Projection of the physical dose objectives. Right: Projection of the LETxD objectives. The plots on the top show the points obtained from the reduction approach for approximation cone $C_{A(\epsilon)}$ using $\epsilon = 0.1$, the plots on the bottom using $\epsilon = 0.01$. The points from the two-step approach are the same in all plots. The orange (two-step) and cyan (SLSVR) points in the top row are the plans used for further comparative analyses.

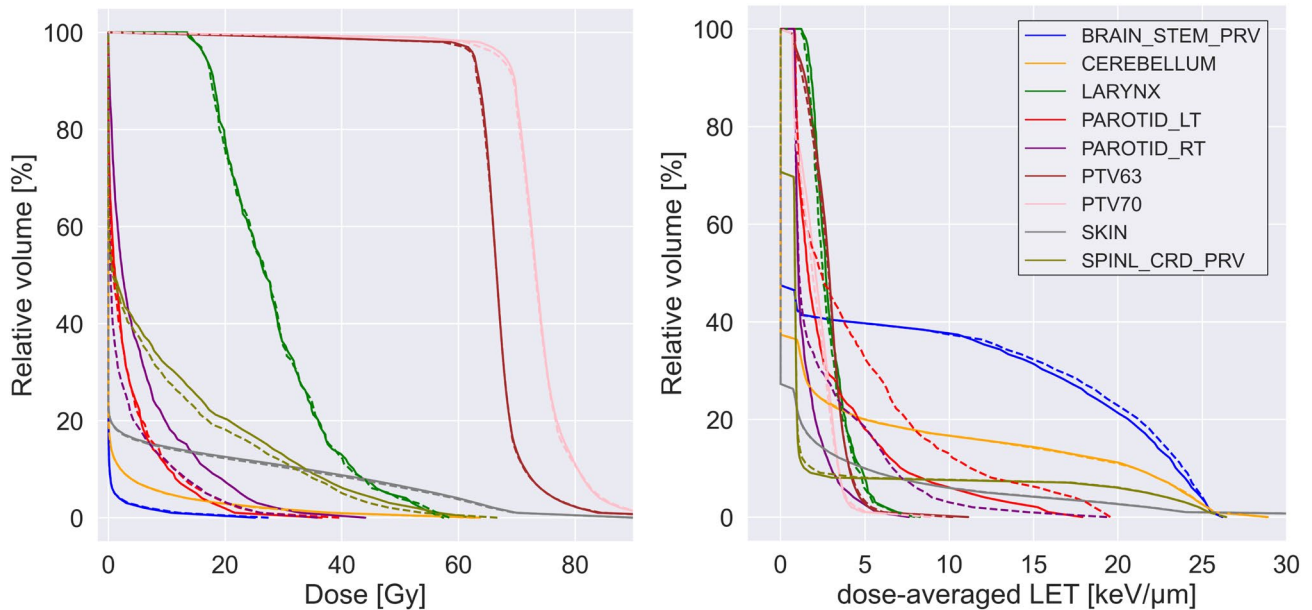


FIGURE 7 | Comparison of two similar optimised plans for the head and neck case, one obtained from the SLSVR reduction approach (highlighted in cyan in Figure 6) and one from the 2-step approach (highlighted in orange in Figure 6, shown as dashed here). Left: Comparison of dose volume histograms showing, on the y-axis, the relative volume of a structure that receives at least the amount of dose on the x-axis. Right: Dose-averaged LET volume histogram, showing the relative volume of a structure that exhibits a dose averaged LET of at least the value on the x-axis.

TABLE 3 | Run times for both algorithms and different settings for the approximation cone, each time calculating a front approximation of 25 plans.

	SLSVR $\epsilon = 0.2$	SLSVR $\epsilon = 0.1$	SLSVR $\epsilon = 0.05$	2-step
Prostate	8453.42	8767.76	7111.65	27059.23
Head-neck	9187.43	14186.98	9553.30	53700.04

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A

Proof of Lemma 1

The following lemma is used in the proof of Lemma 1.

Lemma 6. *If $a \in \text{int}(C)$ and $w \in C^* \setminus \{0\}$, then $w^T a > 0$.*

Proof. Assume $w^T a = 0$ and choose a direction $d_i = -w_i$ for some i such that $w_i \neq 0$ and $d_j = 0$ for $j \neq i$. Because $a \in \text{int}(C)$, there is an $\epsilon > 0$ such that $a + \epsilon d \in C$. However,

$$w^T(a + \epsilon d) = w^T a - \epsilon w_i^2 < 0. \quad (\text{A1})$$

This is a contradiction to $w \in C^*$. $\square$

Proof. (Serna 2012) (Lemma 1.23) Assume that for a $w \in C^* \setminus \{0\}$, z is an optimal solution of $\min_x \left\{ \sum_{i=1}^n w_i F_i(x) \mid x \in X \right\}$. Further, assume z is not weakly non-dominated. This means $\exists \tilde{z} \in F(X): \tilde{z} \in z - \text{int}(C)$. Hence, there is a point $a \in \text{int}(C)$ such that $\tilde{z} = z - a$. From the definition of the dual cone and Lemma 6 it follows

$$w^T \tilde{z} = w^T(z - a) = w^T z - \underbrace{w^T a}_{>0} < w^T z,$$

which contradicts the assumption. Likewise, for $\bar{w} \in \text{int}(C^*)$, assume that z is an optimal solution of the weighted sum but not non-dominated. Thus, there is an $b \in C \setminus \{0\}$ and $\tilde{z} \in F(X)$ such that $\tilde{z} = z - b$. Again, this implies $\bar{w}^T \tilde{z} < \bar{w}^T z$ and consequently contradicts the assumption. $\square$